

Round-7 Audit and Response (Claude)

Adopting GPT-5.5’s pair-shape / incidence-tag repair as the canonical statement

Claude (Anthropic, Opus 4.7), prompted by Michael Wessel

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Disclaimer. This document was drafted by Claude (Anthropic), prompted by Michael Wessel. The results have not been independently verified by human domain experts. We invite scrutiny and corrections.

Abstract

GPT-5.5’s round-7 review of Claude’s round-6 internal-coherence pass of the consolidated v2 proof identifies one central architectural defect in Claude’s manuscript: the abstract pair-label function $L_Q : \text{Pos}_Q^2 \rightarrow \mathbf{B}$ with diagonal $L_Q(\pi, \pi) = \text{EQ}$ collapses distinct laps of a blocking cycle into semantic equality, because in a finite regular certificate two fresh occurrences u_i, u_j of a one-state upward cycle share the same abstract position symbol π and would be assigned $\text{lab}(u_i, u_j) = L_Q(\pi, \pi) = \text{EQ}$. This breaks the canonical certificate for $\exists \text{PP}.\top \sqcap \forall \text{PP}.\exists \text{PP}.\top$ and undermines the central blocking-not-equality slogan. We accept that the round-6 architecture cannot be locally patched: the diagonal-EQ collapse is intrinsic to indexing labels by abstract position symbol pairs. We therefore adopt GPT-5.5’s round-7 proof sketch [2] as the new canonical statement; the expanded full-details version [3] is its detailed companion; the alternative parity-tree-automata proof [4] is retained as an independent witness. Claude’s role shifts back to verification and audit.

This response paper (i) summarizes the round-7 architectural move (incidence tags and pair shapes), (ii) maps the six gaps (G1)–(G6) of Claude’s round-6 paper and the five gaps (D1)–(D5) of GPT-5.5’s round-7 review to their resolution in the round-7 sketch, (iii) reports three self-contained Python verifications (`wp7_selfcontained_side_witness.py`, `wp8_round7_blocking_chain.py`, `wp9_round7_split_copies.py`) corroborating the three central stress concepts of the round-7 architecture, and (iv) discusses the no-automata route as the canonical path with the automata route as the independent witness.

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1 Why round-6 cannot be locally patched

GPT-5.5’s round-7 review [1] identifies a single critical defect in Claude’s round-6 manuscript that is independent of the various surface inconsistencies. Round-6 defined the concrete label on every unfolding pair by

$$\text{lab}(u, v) := L_{\mathcal{Q}}(q(u), q(v)),$$

where $q : \Omega(\mathcal{U}(\mathcal{Q})) \rightarrow \text{Pos}_{\mathcal{Q}}$ sends each concrete occurrence to an abstract position symbol and $L_{\mathcal{Q}} : \text{Pos}_{\mathcal{Q}}^2 \rightarrow \mathbf{B}$ is the declared abstract pair-label function with the standard reflexivity clause $L_{\mathcal{Q}}(\pi, \pi) = \text{EQ}$.

The collapse. In a finite regular certificate, the abstract position alphabet is finite while the unfolding is infinite. Blocking deliberately reuses the same abstract position symbol on infinitely many fresh semantic occurrences. Take the canonical one-state upward cycle for $C_{\uparrow} := \exists \text{PP}.\top \sqcap \forall \text{PP}.\exists \text{PP}.\top$, whose unfolding is

$$u_0 \text{PP} u_1 \text{PP} u_2 \text{PP} \dots$$

with all u_i at the same finite profile, port, context, and stratum. Then $q(u_i) = q(u_j) = \pi$ for every i, j , and the round-6 rule gives

$$\text{lab}(u_i, u_j) = L_{\mathcal{Q}}(\pi, \pi) = \text{EQ},$$

even for $i \neq j$. Two contradictory conclusions follow. Either the quotient identifies u_i and u_j , collapsing the strict PP chain into a PP-self-loop (which violates strong-EQ semantics on PP); or the quotient does not identify them, in which case two distinct elements of the quotient frame are EQ-related (which also violates strong-EQ semantics). Either way the round-6 labelling does not represent the regular certificate it was designed to represent.

Why no local patch suffices. A weaker version of the diagonal clause does not work. Without $L_{\mathcal{Q}}(\pi, \pi) = \text{EQ}$, the reflexive case $u = v$ has no defined label. With a side clause “ $L_{\mathcal{Q}}(\pi, \pi) = \text{EQ}$ except when the two occurrences are different”, the function is no longer a function of $\text{Pos}_{\mathcal{Q}}^2$ alone:

to evaluate at $(q(u), q(v)) = (\pi, \pi)$ one must also know whether $u = v$, which is exactly the information L_Q over position-symbol pairs cannot carry. The defect is intrinsic to the indexing.

The structural move. GPT-5.5's round-7 fix is to redefine the label function on a different finite domain: *occurrence-sensitive pair shapes*. A pair shape $\alpha(u, v)$ carries an explicit incidence tag $\iota(u, v)$ that distinguishes literal identity ($u = v$, tag **self**) from same-symbol fresh copies (tag **up**, **down**, **side_R**, or **front_R**). The label function is

$$\ell_Q : \text{Pair}_Q \rightarrow \mathbf{B}, \quad \ell_Q(\text{self}) = \text{EQ}, \quad \ell_Q(\text{eq}) = \text{EQ}, \quad \ell_Q(\text{up}) = \text{PP}, \quad \ell_Q(\text{down}) = \text{PPI}, \quad \ell_Q(\text{side}_R) = \ell_Q(\text{front}_R)$$

Reflexivity gives **EQ** only when the incidence tag is **self**; two fresh laps with $\iota(u_i, u_j) = \text{up}$ get label **PP**, never **EQ**.

We accept this fix. It is not a small revision of the round-6 formulation, because the entire round-6 architecture (the abstract position symbol Pos_Q , the (V11) composition table on $\mathcal{T}_Q$, the (SC4') per-position support closure, the round-5 (M6') stratum forest, and the round-6 (V15) cross-pair universal safety) is built around the per-position labelling. Replacing it requires reworking the validity-clause numbering, the patchwork proof, the soundness chain, and the verification artefacts. We adopt the round-7 sketch as the canonical proof manuscript and rebuild the verification layer around the new architecture.

2 The round-7 architecture: incidence tags and pair shapes

We summarize the round-7 architecture from [2] for self-containment.

2.1 Incidence tags

Definition 2.1 (Incidence tags, after [2] Def. 5.3). For two concrete occurrences u, v in the unfolding $\mathcal{U}(Q)$, the *incidence tag* $\iota(u, v) \in \text{Inc}$ takes one of the following finite values:

self	$u = v$ as concrete occurrences
eq	$u \neq v$ and $u \equiv_{\text{EQ}} v$ by explicit equality ports
up	v is a strict equality-aware proper superpart of u
down	v is a strict equality-aware proper part of u
side_R	the pair is named in a side mosaic with label $R \in \mathbf{B}$
front_R	the pair is residual sibling-frontier with $R \in \{\text{DR}, \text{PO}\}$.

The tags **self** and **eq** are *occurrence-sensitive*: repeated laps of a blocking cycle never receive **self** merely because their finite profiles agree, and never receive **eq** unless explicitly linked by the equality-port relation.

2.2 Pair shapes and the label function

Definition 2.2 (Pair shape and label, after [2] Def. 5.4–5.5). A *pair shape* is a tuple

$$\alpha(u, v) = (s_u, p_u, s_v, p_v, \iota(u, v)),$$

with s_u, s_v profiles, $p_u \in \text{Port}(s_u)$, $p_v \in \text{Port}(s_v)$, and $\iota(u, v) \in \text{Inc}$. The finite set of permitted pair shapes is $\text{Pair}_{\mathcal{Q}}$. The label function is

$$\ell_{\mathcal{Q}} : \text{Pair}_{\mathcal{Q}} \rightarrow \mathcal{B},$$

with mandatory values

$$\ell_{\mathcal{Q}}(\text{self}) = \text{EQ}, \quad \ell_{\mathcal{Q}}(\text{eq}) = \text{EQ}, \quad \ell_{\mathcal{Q}}(\text{up}) = \text{PP}, \quad \ell_{\mathcal{Q}}(\text{down}) = \text{PPI}, \quad \ell_{\mathcal{Q}}(\text{side}_R) = \ell_{\mathcal{Q}}(\text{front}_R) = R.$$

(For the **self**/**eq** clauses, the profile/port fields are determined by u and v ; for the others, the tag determines the relation directly.)

Remark 2.3. The diagonal collapse of round-6 is averted by Definition 2.1. Two distinct laps u_i, u_j ($i \neq j$) of a blocking cycle satisfy $u_i \neq u_j$ and (typically) $u_i \not\equiv_{\text{EQ}} u_j$. Therefore $\iota(u_i, u_j) \in \{\text{up}, \text{down}, \text{side}_R, \text{front}_R\}$, never **self** or **eq**. In particular, $\ell_{\mathcal{Q}}(\alpha(u_i, u_j)) \neq \text{EQ}$ for any distinct laps; for the one-state upward cycle, the label is **PP**.

2.3 Triple shapes and composition (V6)

Definition 2.4 (Triple shape, after [2] Def. 5.6). A *triple shape* is a tuple $\Theta = (\alpha_{12}, \alpha_{23}, \alpha_{13})$ of pair shapes satisfying coherence constraints: equality endpoints must collapse the third pair label across $\equiv_{\text{EQ}}$; vertical chains must project to vertical labels at the third edge; etc. The finite set of coherent triple shapes is $\text{Tri}_{\mathcal{Q}}$.

(V6) Triple composition. For every $\Theta = (\alpha_{12}, \alpha_{23}, \alpha_{13}) \in \text{Tri}_{\mathcal{Q}}$,

$$\ell_{\mathcal{Q}}(\alpha_{13}) \in \text{comp}(\ell_{\mathcal{Q}}(\alpha_{12}), \ell_{\mathcal{Q}}(\alpha_{23})).$$

This is the round-7 successor of Claude's round-6 (V11). The difference is that triples are over *pair shapes*, not over position-symbol pairs, so the diagonal-EQ collapse cannot occur in the composition check.

2.4 Validity clauses (V1)–(V12)

The round-7 validity clauses, summarized:

- (V1) Type legality (Hintikka types).
- (V2) Context legality (mosaics satisfy local axioms; local **PP**/**PPI** side labels are verticalized).
- (V3) Pair-shape totality (every pair has exactly one shape).
- (V4) Occurrence-sensitive equality (**self** only for $u = v$; **eq** only for explicit equality-port pairs; no strict vertical pair is **eq**).
- (V5) Typed equality congruence (**eq**-mates have identical types and identical $\ell_{\mathcal{Q}}$ -labels to every shape).
- (V6) Triple composition over $\text{Tri}_{\mathcal{Q}}$ (above).

- (V7) Vertical existential support (PP/PPI): equality-aware strict reachability to a witness profile, with strict-later occurrences on cyclic witnesses.
- (V8) Vertical universal safety (PP/PPI): every strictly reachable target has the witnessed type.
- (V9) Side existential support (DR/PO): named mosaic position.
- (V10) Side universal safety (DR/PO): every named side/frontier target carries the type.
- (V11) Request-closed cycles: every cycle is half-open request-closed; cyclic witnesses are strict later occurrences in the same or next lap.
- (V12) Cycle universal safety: universal safety holds across the entire cyclic order including wrap-around.

Twelve clauses versus Claude’s round-5/round-6 fifteen. The reduction is real: round-5 (V11) and round-6 (V15) are absorbed by the round-7 (V6); round-5 (V12)/(V13)/(V14) are absorbed by the new (V9)/(V5)/(V11) respectively (joint equality is folded into typed congruence, declared overlap labels into mosaic legality, uniform cycle representatives into the request-closed-cycle clause directly).

3 Mapping round-6 (G1)–(G6) gaps to round-7 resolutions

GPT-5.5’s round-6 review listed six internal-coherence gaps in Claude’s round-6 consolidated paper. All six are resolved by the round-7 architectural move.

G1 (flat side frontier). Round-6 inlined (M6′) into the Mosaic definition but left the round-4 “Side-interface legality” lemma asserting flat-frontier $\{DR, PO, EQ\}$ on sibling positions. Round-7 ((V2), Mosaic axiom (M6)) requires that any PP/PPI side label live inside a declared local vertical stratum; residual frontier pairs are then restricted to $\{DR, PO\}$. The flat-frontier lemma is dropped.

G2 (SC4′ per-position vs. per-triple). Round-6 weakened (SC4) to per-position but kept Quotient lifting and Patchwork invoking per-triple mosaic coverage. Round-7’s per-position support closure is consistent because composition is checked globally on the triple-shape set Tri_Q via (V6), and patching is by the occurrence-sensitive triple-shape table ([2] Lemma 4.3), not by a per-triple mosaic.

G3 (cross-pair universal safety). Round-6 added (V15) to fix the under-proved universal-propagation lemma. Round-7 folds this into (V5) (typed equality congruence on mosaic positions) and (V8)/(V10) (universal safety on vertical and side targets). Cross-pair universal safety is now a corollary of the universal-safety clauses on the cycle and on the support mosaics.

G4 (extended-profile lasso, half-open cycle). Round-7 adopts the half-open $[i, j)$ convention throughout and uses the extended profile (including pending requests) for the request-closed cycle lemma. No tension between Π and $\hat{\Pi}$ remains.

G5 (bound consistency, V1–V15, stratum-forest grammar). Round-7 has only 12 validity clauses and does not assert a polynomial side-width. The grammar enumerates over occurrence-sensitive pair shapes and triple shapes; the bound on mosaic size is a crude polynomial in $|\text{cl}(C_0)|$ ($p(n) = 1 + 4n + 3(4n + 1)^3$ in [2]), sufficient for decidability. Inconsistency between cubic and recurrence bounds is moot.

G6 (self-contained verification script). Round-7 verification: see §5 below. Three pure-stdlib scripts (`wp7_selfcontained_side_witness.py`, `wp8_round7_blocking_chain.py`, `wp9_round7_split_copies.py`) replace the project-imports-dependent original.

4 Mapping round-7 (D1)–(D5) defects to their resolutions

GPT-5.5’s round-7 review of Claude’s round-6 paper lists, in addition to the central defect (§1), five further issues. All are addressed by the round-7 architectural move.

D1 (equality-port lift ambiguous). Round-6 wrote $u \equiv_{\text{EQ}} v$ iff $\text{Port}(u) \equiv_{\text{EQ}}^Q \text{Port}(v)$ in the same generated context or transitively across explicit links. Read literally, this identifies every occurrence using the same finite port. Round-7 treats explicit equality-port links as *occurrence-template* data generating instantiated equality edges inside context instances; the lift is $u \equiv_{\text{EQ}} v$ iff $u = v$ or there is a path of instantiated explicit equality edges between u and v . Reflexivity at the finite port does not collapse fresh blocked copies.

D2 (V9.b at the finite state-port level). Round-7 restates V9.b at the occurrence-template level: it forbids explicit equality between two positions that can be strict-vertical comparable *in the same unfolding instance*, without rejecting self-loop blocking profiles by reflexivity.

D3 (stratum identity in abstract positions). Round-6 kept a binary $\zeta \in \{\text{vert}, \text{sib}\}$ field on abstract positions, insufficient for nested strata. Round-7’s mosaic declares vertical strata as a finite family $\mathcal{V}$, and the co-vertical pair shape carries the stratum identity directly; the binary flag is dropped.

D4 (recursive side-interface underspecified for chains). Round-6’s recursive side-interface construction was ambiguous for side-witness chains $w_0 \text{PP} w_1 \text{PP} w_2$. Round-7 (M6) is the simpler condition: every PP/PPI side label in a mosaic lies in a declared vertical stratum; the stratum carries an explicit local containment order, so chains are represented as ordered sequences inside a single declared stratum.

D5 (universal wrap-around in pumped cycles). Round-6’s request-cycle proof preserved existentials but the wrap-around case of universals was under-proved. Round-7 adds (V12) Cycle universal safety: every $\forall R.D$ at any cycle position holds at every later cycle position including wrap-around. This is a finite cycle-word check.

5 Verification corroboration

We supply three self-contained Python verifications, each in `verification/python/`, that exercise the three central stress concepts of the round-7 architecture. Each script uses only the Python standard library and runs from the verification archive alone.

5.1 WP7: side-witness comparability

`wp7_selfcontained_side_witness.py` (round-6 (G6) repair) builds the three-element abstract RCC5 frame $\Delta = \{x, y, z\}$ with $\rho(x, y) = \text{DR}$, $\rho(x, z) = \text{DR}$, $\rho(y, z) = \text{PP}$, $A = \{y\}$, $B = \{z\}$, and verifies:

- JEPD, inverse, strong-EQ, and the full RCC5 composition table on every pair and triple;
- $C_{\text{side}} := \exists \text{DR}.(A \sqcap \exists \text{PP}.B) \sqcap \exists \text{DR}.B$ is SAT at x .

The script confirms the round-5/round-7 claim that two DR-witnesses of a source can themselves be PP/PPI-comparable. The round-7 architecture handles this via local vertical stratification inside a side mosaic.

5.2 WP8: blocking is not equality

`wp8_round7_blocking_chain.py` (round-7 central repair) truncates the canonical one-state upward cycle of $C_{\uparrow} := \exists \text{PP}.\top \sqcap \forall \text{PP}.\exists \text{PP}.\top$ to a five-element frame $\{u_0, \dots, u_4\}$ with $\rho(u_i, u_j) = \text{PP}$ for $i < j$, PPI for $i > j$, EQ for $i = j$. The script verifies:

- JEPD, inverse, strong-EQ, RCC5 composition (frame conditions);
- the round-7 incidence-tagged label rule keeps every distinct lap pair (u_i, u_j) at label PP or PPI, *never* EQ;
- the round-6 collapsed rule $\text{lab}(u, v) = L_Q(q(u), q(v)) = L_Q(\pi, \pi) = \text{EQ}$ does mis-equate distinct laps (explicit demonstration of the defect).

$C_{\uparrow}$ has no finite model, so SAT is not asserted at the truncation; the script exercises the architectural claim about the regular certificate.

5.3 WP9: split copies for incomparable proper superparts

`wp9_round7_split_copies.py` builds the three-element abstract RCC5 frame $\Delta = \{x, a_{\text{sup}}, b_{\text{sup}}\}$ with $\rho(x, a_{\text{sup}}) = \text{PP}$, $\rho(x, b_{\text{sup}}) = \text{PP}$, $\rho(a_{\text{sup}}, b_{\text{sup}}) = \text{PO}$ (forced by strong-EQ semantics: a_{sup} and b_{sup} both contain x , hence share a point, hence PO), $A = \{a_{\text{sup}}\}$, $B = \{b_{\text{sup}}\}$, and verifies:

- the quotient frame respects JEPD, inverse, strong-EQ, and RCC5 composition;
- the pre-quotient occurrence structure $(x, x_a, x_b, a_{\text{sup}}, b_{\text{sup}})$ with explicit equality-port chain $x \text{eq}_a \text{eq}_b$ satisfies the round-7 typed equality congruence (the three mates have identical ℓ_Q -labels to every outside position);
- no distinct pair receives the **self** tag (round-7 no-self-collapse);
- $C_{\text{split}} := \exists \text{PP}.A \sqcap \exists \text{PP}.B$ is SAT at x .

This exercises the round-7 split-copy architecture for multiple selected proper superparts.

5.4 Summary

All three scripts pass. The empirical layer corroborates the round-7 architectural claims for the three canonical stress concepts without relying on any project-internal Python module.

6 No-automata route vs. automata route

GPT-5.5 has supplied two round-7 proofs of the same decidability theorem.

No-automata route (canonical). [2] (14 pages) and the expanded companion [3] (28 pages) use direct finite-index blocking with explicit request-closed cycles. No Büchi or parity automata appear. This route is the canonical statement because it stays inside the same conceptual layer as the certificate validity check: everything is a finite syntactic test on finite alphabets.

Automata route (independent witness). [4] (21 pages) reduces satisfiability to non-emptiness of a parity tree automaton over the split-forest alphabet. This route is an independent reproof of the decidability theorem using a different finite-state machinery. It is kept in the repository as a corroboration of the no-automata route, not as the canonical statement.

Why this matters. The two routes converge on the same theorem (concept satisfiability of $\mathcal{ALCI}_{RCC5}$ over strong abstract atomic $RCC5$ frames is decidable). The no-automata route is a more direct argument (ω -regularity is replaced by a finite combinatorial cycle-closure condition) and aligns with the verification layer’s preference for finite enumeration. The automata route is a classically-styled witness: if a reviewer wants to translate the proof to a familiar automata-theoretic setting, the parity-automaton construction is already available.

7 Status of Claude-authored manuscripts

The following Claude-authored manuscripts are now superseded by GPT-5.5’s round-7 sketch and expanded companion:

- `papers/gpt5.5_round2/repaired_split_forest_no_automata_proof_v2.tex` (May 2026, 15 pages, the round-5 (M6’) delta);
- `papers/gpt5.5_round2/repaired_split_forest_no_automata_proof_items4to8.tex` (May 2026, 12 pages, the round-5 items 4–8 delta);
- `papers/gpt5.5_round2/repaired_split_forest_no_automata_proof_v2_consolidated.tex` (May 2026, 36 pages, the round-5 consolidation);
- `papers/gpt5.5_round2/repaired_split_forest_no_automata_proof_v2_consolidated_round6.tex` (May 2026, 39 pages, the round-6 internal-coherence pass).

All four are retained in the repository as the historical audit trail. The conceptual contributions that survive into round-7 are:

1. the recognition that two DR/PO side witnesses of the same source can be PP/PPI-comparable (round-5 stress concept, now worked example in [2] §6 and verified in §5.1); and
2. the broader two-layer architecture (theoretical proof + self-contained executable verification artefacts) that surfaced the round-6 defect.

The round-7 sketch is the new canonical pillar of the decidability argument. Claude’s role going forward is verification, audit, and documentation, not authorship of the proof manuscript.

8 Conclusion

GPT-5.5’s round-7 architectural move (occurrence-sensitive pair shapes with incidence tags) resolves the central defect of Claude’s round-6 manuscript and absorbs the smaller round-6 (G1)–(G6) and round-7 (D1)–(D5) issues. The three self-contained verification scripts (WP7, WP8, WP9) corroborate the three central stress concepts of the round-7 architecture. We adopt GPT-5.5’s round-7 sketch as the new canonical statement; the expanded full-details version is its detailed companion; the alternative automata-route proof is retained as an independent witness. Claude’s previous round-5 and round-6 manuscripts are marked superseded and preserved as the audit trail.

References

- [1] GPT-5.5 (OpenAI), prompted by Michael Wessel. *Formal Review of the Round-6 Consolidated Generated Split-Forest Proof*. Unpublished manuscript, May 27, 2026. Repository: https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5_final/formal_response_round6_review.pdf.
- [2] GPT-5.5 (OpenAI), prompted by Michael Wessel. *A Repaired Split-Forest Decidability Proof for $\mathcal{ALCI}_{RCC5}$ Without Automata*. Unpublished manuscript (consolidated round-7, 14 pages), May 2026. Repository: https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5_final/repaired_split_forest_all_in_one_round7.pdf.
- [3] GPT-5.5 (OpenAI), prompted by Michael Wessel. *Expanded Generated Split-Forest Decidability Proof for $\mathcal{ALCI}_{RCC5}$: Detailed Companion*. Unpublished manuscript (round-7, 28 pages), May 2026. Repository: https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5_final/expanded_split_forest_full_details_proof.pdf.
- [4] GPT-5.5 (OpenAI), prompted by Michael Wessel. *A Split-Forest Decidability Proof for $\mathcal{ALCI}_{RCC5}$ via Parity Tree Automata: Detailed Version*. Unpublished manuscript (round-7, 21 pages), May 2026. Repository: https://github.com/lambdamikel/alcircc5/blob/master/papers/gpt5.5_final/split_forest_automata_decidability_proof_detailed.pdf.
- [5] Michael Wessel et al. *ALCI_RCC5 Project Repository*. <https://github.com/lambdamikel/alcircc5>.